

Driver-Based Multivariate Time Series Forecasting

A Comparative Analysis of Meta-Learning vs Ensemble
Performances

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Introduction & Context

Context & Problem

- **Goal:** Predict "Driven" target using "Drivers" (influencing variables) in domains like Environmental monitoring, meteorology and Finance.
- **Industrial Origin:** Developed with industry partners; tested on public data as proprietary datasets were confidential.
- **Problem:** Linear models (ARIMA) fail with non-linear dependencies; Ensemble methods are computationally expensive; Manual tuning is inefficient and prone to error.

Proposed Solution: Meta-Learning

We propose a Meta-Learning approach to predict optimal forecasting parameters based on data characteristics.

- **Prediction:** Automatically selects Model, Window Size, and Feature Selection.
- **Benefits:** Improves generalization, reduces manual tuning, lowers computational cost, and mitigates overfitting on short history.

Methodology: Dataset

Air Quality Monitoring in European Cities

-  **Source:** Public dataset (Angelis et al., 2024), covering 2020-2023 with hourly resolution.
-  **Drivers:**
 - Pollutants: Ozone (O₃), nitrogen dioxide (NO₂) and particulate matter (PM).
 - Meteo: Wind Speed (Zonal U/Meridional V), Temp (°C), total precipitation, relative Humidity.
-  **Target:** O₃ concentration (selected via feature selection due to clear meteorological dependence).



Methodology: Benchmarking Setup

Time series grouped in bins of 50 time series by length based on periodicity (~24h, informed by ACF/PACF).



Mini

0 – 24 hours

Very short
history



Small

24 – 48 hours

1-2 Days



Medium

48 – 96 hours

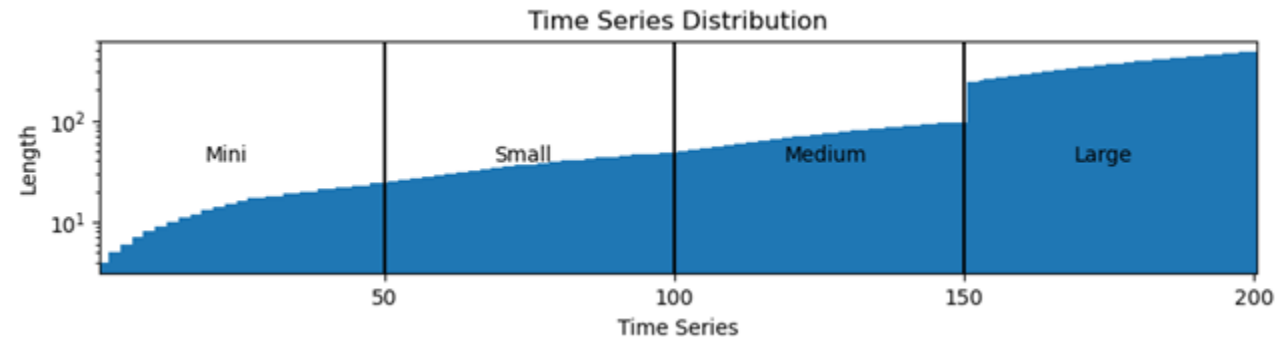
2-4
Days



Large

240 – 480 hours

10-20 Days



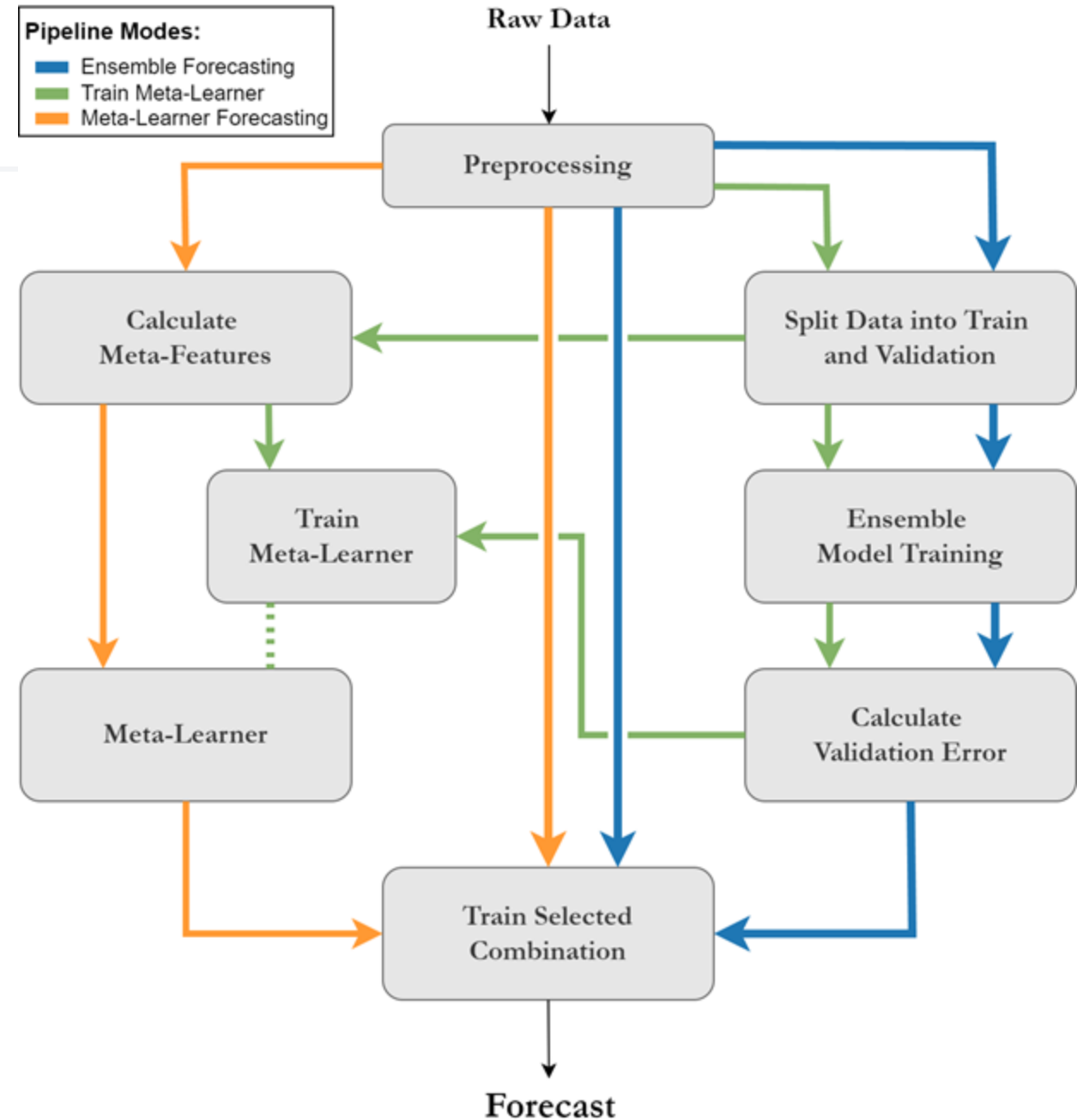
System Implementation

Modular Workflow

A unified pipeline sharing preprocessing steps across three execution modes:

- 1 **Ensemble Forecasting:** Exhaustive search on validation sets.
- 2 **Train Meta-Learner:** Building the prediction model.
- 3 **Meta-Learner Forecasting:** Inference mode for new series.

MBSE Integration: Artifacts (models, configs) allow integration with Model-Based Systems Engineering toolchains for traceability and "what-if" analysis.



Ensemble vs. Meta-Learner Strategies

Feature	Ensemble Forecasting	Meta-Learner Forecasting
Approach	Exhaustive Search	Predictive Selection
Validation	On validation set matching horizon	Predicted by Neural Network
Cost	High (trains many models)	Low (inference + single train)
Input	Raw Time Series	Meta-Features (Stats, Trend, etc.)
Adaptability	High (brute force)	High (learned generalization)

Meta-Learner Architecture

Neural Network Design

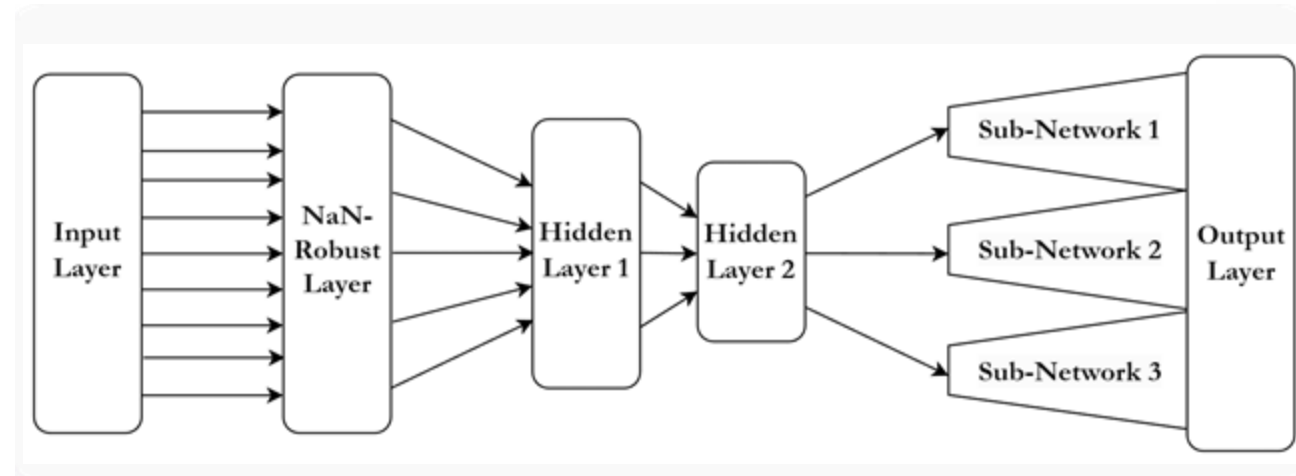
➔ **Input tensor:** driver and driven Meta-features (Mean, Std, Skew, Trend, Periodicity via spectral analysis).

🔧 **Core:** Neural Network equipped with a **NaN-Robust Layer** to handle missing sensor data without simple imputation.

🌐 **Output Multi-Head tensor:**
SN1: Feature Selection Strategy

SN2: Window Size Class

SN3: Forecasting Model Type





Input Tensor Meta-Features

For Each Driver

Characterizing the input variables:

- Mean
- Standard deviation
- Skewness
- Kurtosis
- Correlation with target

For Target Series

Characterizing the "Driven" variable:

- Length of time series
- Dominant period
- Trend strength
- Seasonality strength
- Residual std dev & Outlier count

Tensor Example:

- `X= [ D1_Mean, D1_Std, D1_Corr, ..., D2_Mean, D2_Std, D2_Corr, ..., T_Trend, T_Seasonality ]`
 - **Positions 0-4:** Are *always* reserved for Driver 1.
 - **Positions 5-9:** Are *always* reserved for Driver 2.
 - **Final Positions:** Are *always* reserved for the Target (Driven).



Output tensor Forecasting Parameters

Parameter	Categories / Options
Input Window Size	Mini (0-14h), Small (24-48h), Medium (48-96h), Large (240+h)
Feature Selection	None, VIF (Variance Inflation Factor), SHAP, Univariate
Forecasting Model	AutoARIMA, Lasso, Random Forest, XGBoost, MLP, LSTM
Validation Metric	MAE, MASE

Tensor Example

$Y = \{Y_Window, Y_Selection, Y_Model\}$

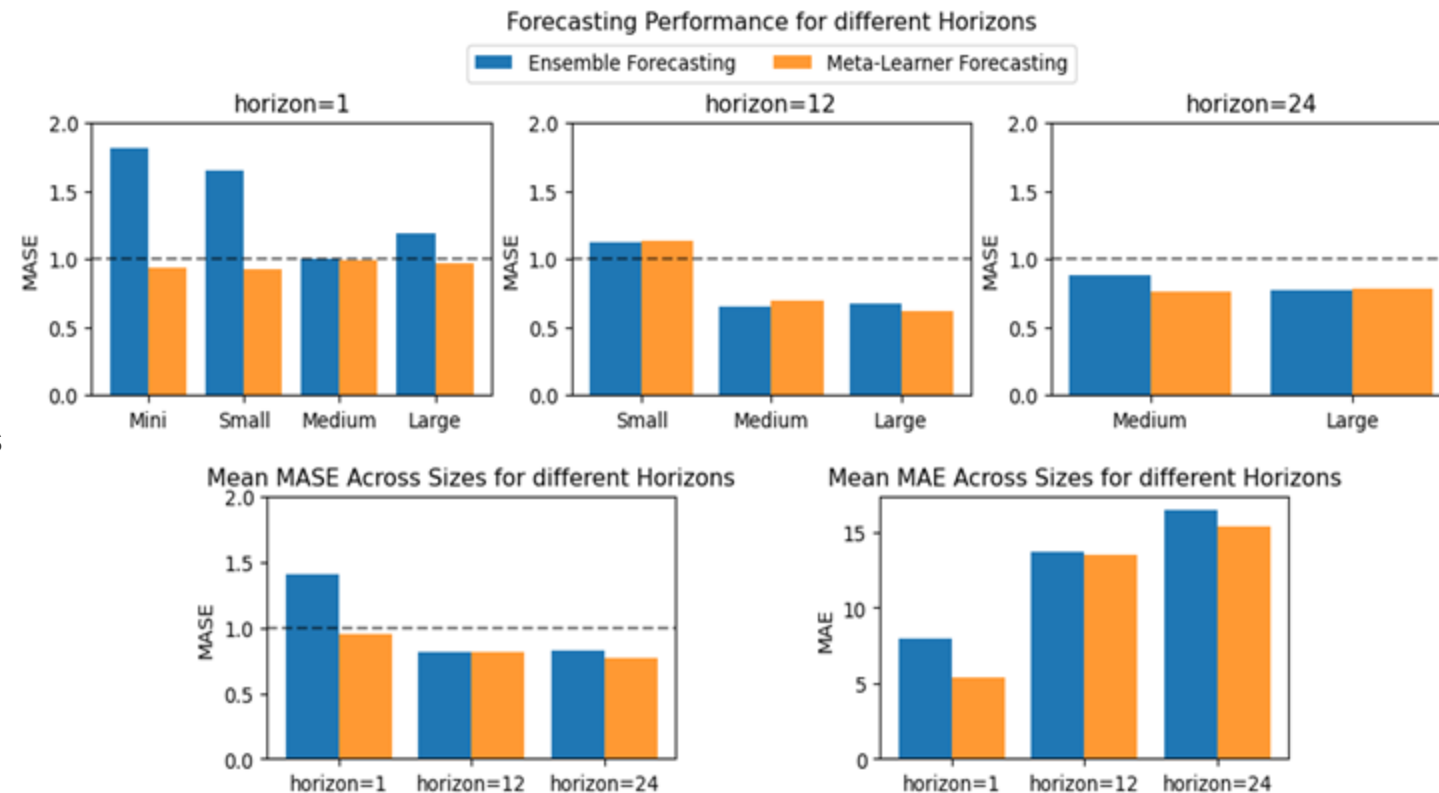
- **Y_Window** : A probability vector over the window classes.
 - Example: $[0.1, 0.8, 0.1, 0.0]$ (Meaning: 80% confidence that "Small" is the best window size).
- **$Y_Selection$** : A probability vector over feature selection strategies.
 - Example: $[0.05, 0.05, 0.9, 0.0]$ (Meaning: 90% confidence that "SHAP" is the best strategy).
- **Y_Model** : A probability vector over the forecasting algorithms.
 - Example: $[0.0, 0.2, 0.0, 0.7, 0.1, 0.0]$ (Meaning: 70% confidence that "XGBoost" is the best model)

Results: Forecasting Performance

Performance Across Horizons

Metrics: MASE (Mean Absolute Scaled Error) & MAE.

- Horizon 1:** Meta-Learner clearly outperforms Ensemble. Even for Mini/Small datasets, it achieves $MASE < 1$.
- Horizon 12 & 24:** The performance gap narrows as the horizon increases, but the Meta-Learner remains highly competitive while being significantly more efficient.



Results: Impact of Data Size

Small vs. Large Datasets



Small Data: Meta-Learner is significantly better.

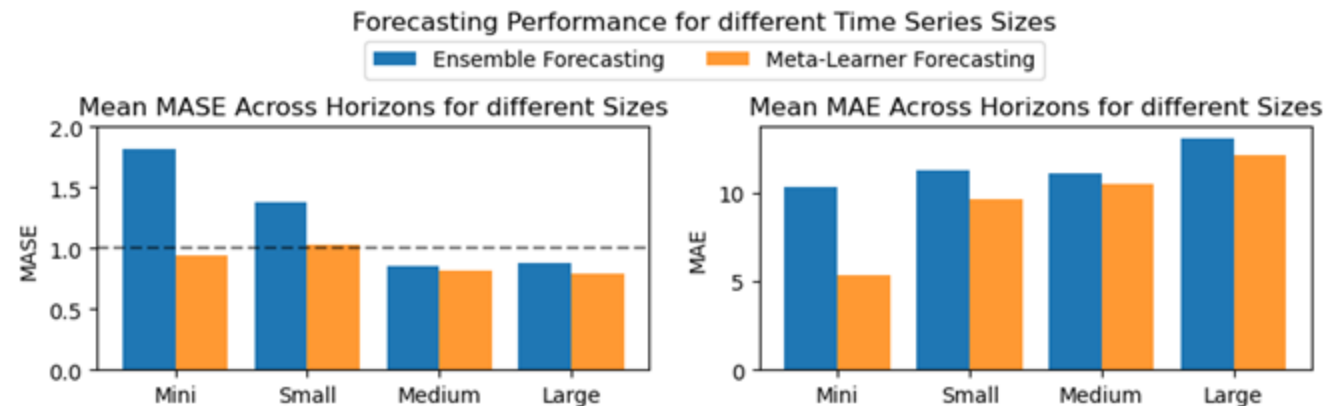
Ensemble strategies struggle here because splitting small data for validation leaves insufficient training samples.



Large Data: The Ensemble method improves as data volume grows, eventually catching up to the Meta-Learner's accuracy.



Adaptability: Meta-Learner correctly identifies when to use simpler models for shorter/noisier series.



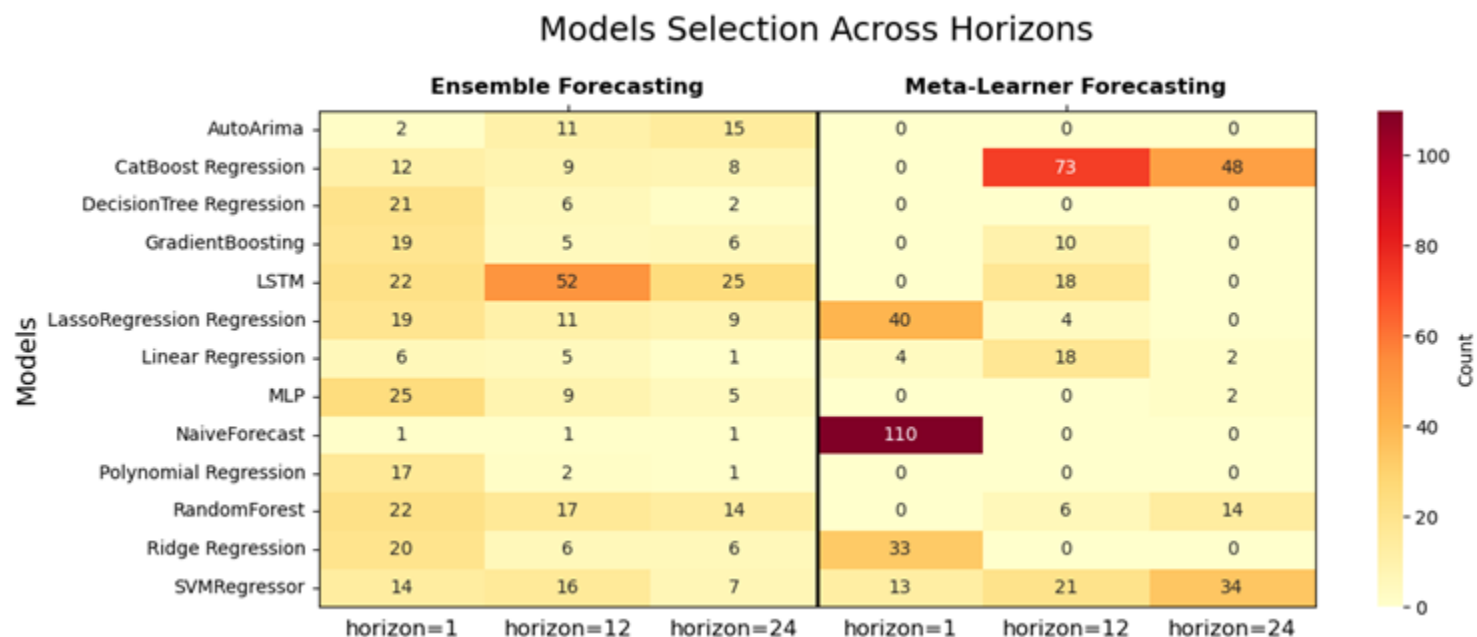
Results: Model Selection Patterns

Ensemble Strategy

- **Horizon 1:** Favors complex models (LSTM, MLP, Decision Trees) for smaller horizons.
- **Longer Horizons:** reduces somewhat the complexity of the models (Random Forest, Autoarima, Lasso)

Meta-Learner Strategy

- **Horizon 1:** Highly adaptive; frequently selects Naive or Linear models where appropriate.
- **Longer Horizons:** Increases complexity (CatBoost, Lasso) only when data supports it.



Both approaches converge towards simpler models (decision trees) as the forecasting horizon increases.

Discussion & Key Findings



Efficiency

Drastically reduces computation by eliminating the need to train 50+ candidate models per series.



Accuracy

Generalizes better on "Mini" and "Small" datasets where traditional validation splits fail.



Adaptability

Successfully captures the non-linear relationship between meta-features and optimal configurations.

Limitation: Currently single-domain (Air Quality) and industrial use case (Cashflow). Future work to expand to Energy/Finance.

Conclusion

- ✓ **Superior Performance:** Meta-Learner outperforms Ensemble Forecasting for short horizons and small historical datasets.
- ★ **Key Contribution:** Extended meta-learning to predict **Window Size** and **Feature Selection**, not just the Model type.
- ⚠ **Future Work:**
 - Conduct ablation studies to isolate feature impacts.
 - Test applicability on heterogeneous domains (Finance, Energy).
 - Explore transformer-based architectures for the Meta-Learner.

Q&A

Thank you for your attention.

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**DTx -- Digital Transformation
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This work was supported under the base funding project of the DTx CoLAB - Collaborative Laboratory, under the **Missão Interface of the Recovery and Resilience Plan (PRR)**, integrated in the notice 01/C05-i02/2022. We thank our Portuguese industry partners for their collaboration.